

Nathan Hammerschmitt Le Gal

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Education

Dartmouth College, Hanover, NH

Sep 2023 – Present | GPA: 3.82/4.0

Bachelor of Arts: Engineering Sciences, Studio Art Modifier

Bachelor of Engineering: Mechanical Engineering

Selected Coursework: CAD & Kinematics, Solid Mechanics, Fluid Mechanics, Materials Science, Analog Circuits, Systems, Distributed Systems

Work Experience

Neros Technologies, *Incoming Mechanical Engineering Intern*

Knoxville, TN | Jun 2026 – Present

- Designing operator interface hardware systems, supporting CMs, integrating DFM & rapid prototyping

Nexus, *Mechanical Engineer & Co-Founder*

Hanover, NH | Jan 2025 – Present

- Created end to end workflow for custom-fit, FDM printed shoes made on-demand using Volumetric AI foot scans, OBJ adjustment scripts, and pre-designed Blender mesh files
- Produced 50+ design-test iterations varying upper wall thickness, midsole thickness, overlay layout, and support placement to reduce print time 26% and material usage 11%
- Improved surface quality of foaming TPU through a 22% reduction in travel movements, testing 6+ extrusion temperatures, and constructing an overhead filament feeding system to minimize friction
- Scaled to 80 units/month with 6 Prusa XLs and automated g-code generation using headless PrusaSlicer
- Secured \$40k in venture and grant funding by communicating technical milestones and validation goals

UMass Amherst Locomotion Lab, *Biomechanics Research Assistant*

Amherst, MA | Jun 2024 – Aug 2024

- Analyzed accelerometer data with 20+ python scripts using Butterworth filters, dynamic time warping, and normalization to derive angular velocity based metrics of repeatability and symmetry of outdoor trail running
- Assisted PhDs and post-docs in motion-capture, IMU, and metabolic data collections for published studies

Projects

Hydrogen Electrocatalyst Lattice Design, *Materials Science*

Jan 2026

- Led 5-person team to fabricate cobalt microlattices using polymer infusion additive manufacturing (PIAM) and varied calcination holding temperature to reduce catalyst Tafel slope from 85 to 70 mV/dec
- Analyzed and tested samples to assess reaction kinetics and structural features using LSV, XRD, and SEM

Truss Bridge, *Solid Mechanics*

Aug 2025

- Led 3-person team to design and fabricate composite plywood bridge (0.84 kg) sustaining 20 kN under Instron testing, exceeding simulated FOS of 1 under 16 kN
- Optimized truss geometry using SolidWorks FEA and SAP2000 by reducing non-critical member width, reinforcing load-bearing members, and selecting 170 mm height for strength-to-weight efficiency

Dino Robot Walker, *Computer Aided Design and Kinematics*

May 2025

- Designed radio-controlled bipedal robot with a 4-bar linkage using Geogebra and Grashof analysis
- Modeled and assembled system in SolidWorks using laser cut and 3D-printed components, lock nuts, heat inserts, washers, screws, and servos to produce final assembly drawings and bill of materials

Skills

CAD & Design: SolidWorks (CSWA), Fusion 360, SAP2000, Blender, GD&T, technical drawings

Fabrication: FDM 3D-printing, Trak DPM Bed Mill, laser cutting, DFM & DFA

Programming: Python, C, MATLAB, Arduino, Git

Additional Information

Leadership: Dartmouth Triathlon & Cycling Team Equipment Executive, Magnuson Center Leadership Board

Accomplishments: 4 Citations for Meritorious performance, Leapyear Startup Accelerator (\$30k), Henry David Thoreau Scholarship (\$25k), CST Scholarship (\$10k), Agganis Scholarship (\$8k), High School Valedictorian

Languages: English, French